

Geometry
Unit 10
Lesson 8 Practice

Name Answer Key
Date _____
Period _____

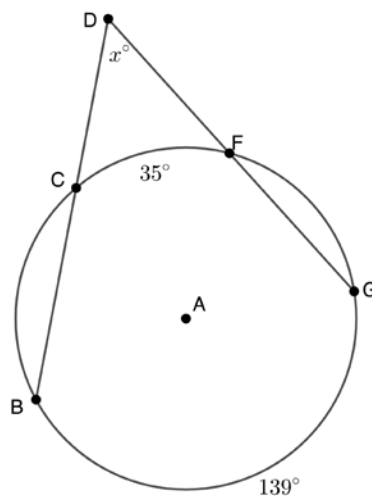
For questions 1-2, Circle A has points B , C , F , and G on the circle. Secant lines $\overline{BD}$ and $\overline{GD}$ intersect at point D outside the circle. The $m\widehat{BG} = 139^\circ$, $m\widehat{CF} = 35^\circ$, and $m\angle CDF = x^\circ$.

1. Write a formula that can be used to find the value of x .

$$x = \frac{139-35}{2}$$

2. Find the value of x .

$$x = \frac{139-35}{2} = 52$$



For questions 3-5, Circle K has points N , H , L , and M on the circle. Secant lines $\overline{NJ}$ and $\overline{MJ}$ intersect at point J outside the circle. The $m\widehat{LH} = (x + 12)^\circ$, $m\widehat{MN} = (3x - 28)^\circ$, and $m\angle HJL = 28^\circ$.

3. Find the value of x .

$$\frac{(3x-28)-(x+12)}{2} = 28$$

$$2x - 40 = 56$$

$$2x = 96$$

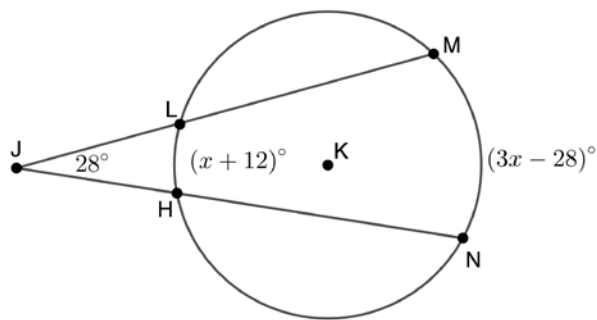
$$x = 48$$

4. Find the $m\widehat{MN}$.

$$m\widehat{MN} = 3(48) - 28 = 116^\circ$$

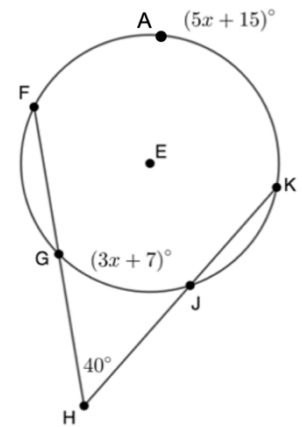
5. Find the $m\widehat{LH}$.

$$m\widehat{LH} = 48 + 12 = 60^\circ$$



Not drawn to scale

For questions 6-8, Circle E has points F , G , J , and K on the circle. Secant lines $\overline{FH}$ and $\overline{KH}$ intersect at point H outside the circle. The $m\widehat{FAK} = (5x + 15)^\circ$, $m\widehat{GJ} = (3x + 7)^\circ$, and $m\angle GHJ = 40^\circ$.



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6. Find the value of x .

$$\frac{(5x+15)-(3x+7)}{2} = 40$$

$$2x + 8 = 80$$

$$2x = 72$$

$$x = 36$$

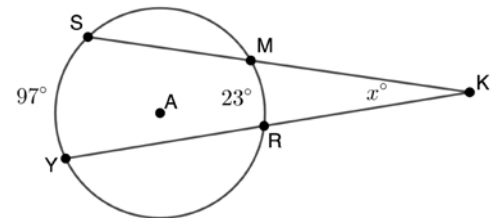
7. Find the $m\widehat{GJ}$.

$$m\widehat{GJ} = 3(36) + 7 = 115^\circ$$

8. Find the $m\widehat{FAK}$.

$$m\widehat{FAK} = 5(36) + 15 = 195^\circ$$

For questions 9-10, Circle A has points S , M , Y , and R on the circle. Secant lines $\overline{SK}$ and $\overline{YK}$ intersect at point K outside the circle. The $m\widehat{SY} = 97^\circ$, $m\widehat{MR} = 23^\circ$, and $m\angle MKR = x^\circ$.



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9. Write a formula that can be used to find the value of x .

$$\frac{97-23}{2} = x$$

10. Find the value of x .

$$x = \frac{97-23}{2} = 37$$